Data Science Project

Census

01. INTRODUCTION

This report is on a mock census of an imaginary modest town located between two larger cities. Hypothetically, I am part of a local government team who will be making decisions on what to do with an unoccupied plot of land and what to invest in depending on population growth projections, availability of resources, demand and other socio-economic considerations. My recommendations will be based on careful analysis of the data provided.

1.0 DATA CLEANING

This was done column by column to make sure each single cell is checked for structural errors and/or missing data, Structural errors were fixed on a case by case basis, then missing values were corrected by inputting values based on actual observations from the data. Every cleaning procedure is detailed comprehensively and the history provided in the attached Jupyter Notebook.

1.1 Data Information

Having read the dataset to pandas DataFrame as df, we viewed the dataset with df.head(3) to understand the basics, and general layout. The dataset is made up of 11 columns as shown below;

	House Number	Street	First Name	Surname	Age	Relationship to Head of House	Marital Status	Gender	Occupation	Infirmity	Religion
0	1	Regentchain Avenue	Ashleigh	Dale	41	Head	Divorced	Female	Field trials officer	None	Catholic
1	1	Regentchain Avenue	Deborah	Dale	11	Daughter	NaN	Female	Student	None	NaN
2	1	Regentchain Avenue	Gary	Dale	7	Son	NaN	Male	Student	None	NaN

The dataset was also inspected for NaN values, and there were 1975 NaN values in the 'Marital Status' column and 2012 NaN values in the 'Religion' column. This was subsequently replaced with the most appropriate entries based on the given circumstances.

1.2 Column 1: 'House Number' and Column 2: 'Street' have no missing or NaN values.

1.3 Column 3: 'First Name'

Index 1497 was missing a first name, and instead replaced it with the salutation " Mr." instead of fabricating a name. Moreover, the person can easily be identified as "Mr Frost." despite lacking a first name.

1.4 Column 4: 'Surname'

Linda at index 60, a member of the Patel family at 8 Telegraphnet Street, was missing a surname. So I replaced it with the family surname on the assumption that she is sharing the same surname with the rest of her family. Also, Denise at index 4236, a 9 yr-old adopted daughter in the Lloyd's household at 48 Chan Estate lacked a surname and it was also replaced with the family surname.

1.5 Column 5: 'Age'

There was a missing value in column 5: 'Age' and this was replaced with 21, since the resident, Howard Baker is a University Student, we estimated his age to be 21. He is also a sibling to a pair of twins who are both 19. Column 5: 'Age' was also converted from integer to float to enable better analysis.

1.6 Column 6: Relationship to Head of House has no missing or NaN values

1.7 Column 7: 'Marital Status'

Before the data was cleaned, Column 7: 'Marital Status' contained;

- 2999 'Single' entries
- 2224 'Married' entries
- 791 'Divorced' entries
- 1 empty value and,
- 1975 NaN values

The empty cell at index 4230 was replaced with 'Single', this is a 25 year old university student who lives with his parents. The 1975 NaN values turned out to be residents under 18 years of age. Their marital status was subsequently changed to 'Minor/NA'. (NA means 'Not Applicable'). This was based on the fact that the legal age for marriage in the UK is 18+ years.

Marital Status after the data was cleaned

```
In [32]: #Checking 'Marital Status' after cleaning.
         df['Marital Status'].value_counts()

Out[32]: Single      3000
         Married     2224
         Minor/NA    1975
         Divorced     791
         Widowed      387
         Name: Marital Status, dtype: int64
```

1.8 Column 8: 'Gender'

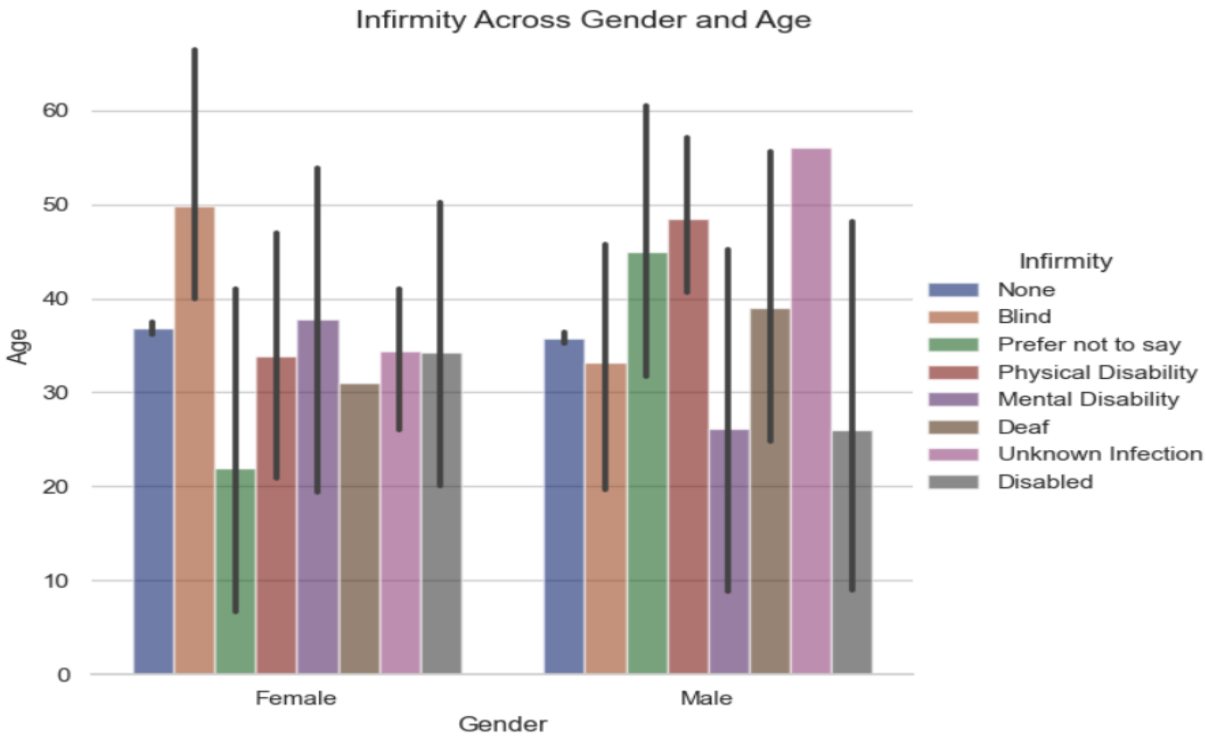
There were 3 rows with missing values in column 8: 'Gender'. At index 1002, there is Pamela Edwards, 69, who is the Head of House. The only other resident in that house is a married male Gary. Therefore, since there are only two genders in the mock Census. It is reasonable to assume that Pamela is the wife of Gary. Same applies to Irene Harvey, 51, at index 7348l. The decision to substitute the absent gender with 'Female' at index 6899 was made solely because the resident has a typical female name. There was no other feasible premise, since the resident has no relationship with the head of her house.

1.9 Column 9: 'Occupation'

Josephine Anderson, 73 at index 3399, has a missing value. Since she is 73, we can assume that she is retired as many individuals retire around that age. However, forced retirement is no longer applicable in the UK.

1.10 Column 10: ‘Infirmity’

The infirmity column has 13 missing values which was replaced with ‘Prefer not to say’. The assumption is that the people that left it blank preferred not to answer due to maybe privacy concerns.



1.11 Column 11: ‘Religion’

The data cleaning was done by replacing 'Jedi', 'Quaker', 'Housekeeper', some blanks, and even 'Private' with 'Prefer not to say' on the supposition that this is what the resident wanted to communicate. The number of unique entries was reduced by merging with Christian, since Quakers is a Christian denomination not a religion.

There are a total of 2012 NaN entries in the ‘Religion’ column. 1978 out of the 2012 where from residents under 18, these were replaced with the religion of the head of the house they live in, on the assumption that it is highly probable that minors will be practicing the same religion with the head of their household. It can be inferred that parents in the UK have considerable influence on the religion of their wards. According to Duncan Lewis Solicitors (n.d.), “Children can legally choose to follow any religion at any age – however, if parents feel that following a specific religion may exert a harmful influence over a child below the age of 18, they can apply to the court for wardship and have their child made a Ward of Court” (para. 4).

After the data was cleaned

In [64]:
```
df['Religion'].value_counts()
```

Out[64]:
```
None                  3515
Christian             2597
Catholic              1209
Methodist              785
Muslim                 119
Prefer not to say       69
Sikh                    42
Jewish                  36
Buddist                  3
Pagan                    1
Orthodoxy                1
Name: Religion, dtype: int64
```

1.11 The dataset infomation after cleaning was concluded

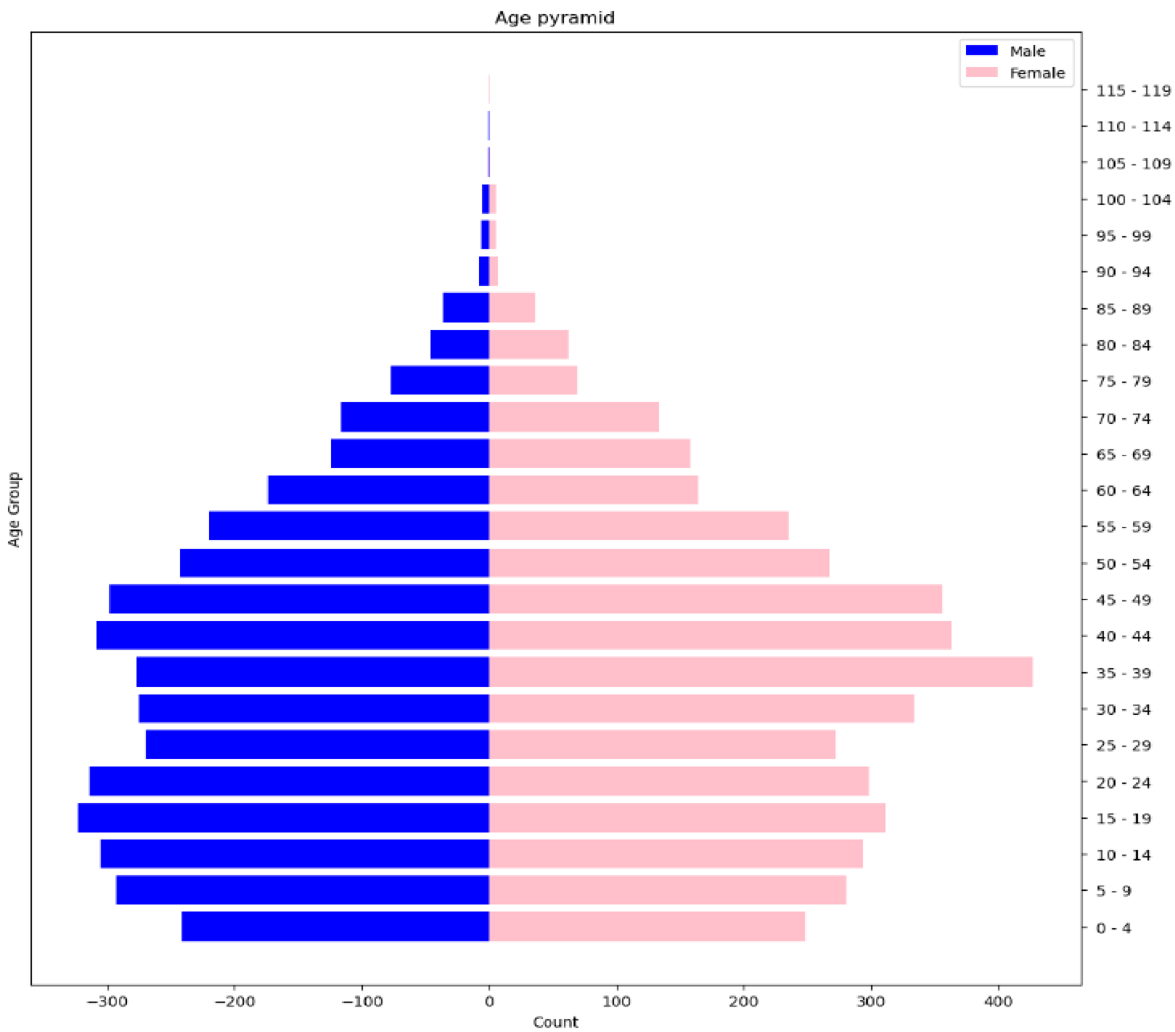
In [71]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8377 entries, 0 to 8376
Data columns (total 12 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   House Number                   8377 non-null   int64
 1   Street                         8377 non-null   object
 2   First Name                     8377 non-null   object
 3   Surname                        8377 non-null   object
 4   Age                            8377 non-null   int32
 5   Relationship to Head of House  8377 non-null   object
 6   Marital Status                 8377 non-null   object
 7   Gender                         8377 non-null   object
 8   Occupation                     8377 non-null   object
 9   Infirmity                      8377 non-null   object
 10  Religion                       8377 non-null   object
 11  Age Group                      8288 non-null   category
dtypes: category(1), int32(1), int64(1), object(9)
memory usage: 696.4+ KB
```

2.0 ANALYSIS AND VISUALIZATION

2.1 Examining the Age Distribution of the Town's Population

In this section, answers to the questions such as; Is the population growing or shrinking? Will there be more retired aged people in the future, more school-aged children, more young people, will be provided using visualizations and other techniques. The analysis was carried out mainly with the age pyramid, as shown below.



A. Interpreting the Population Pyramid

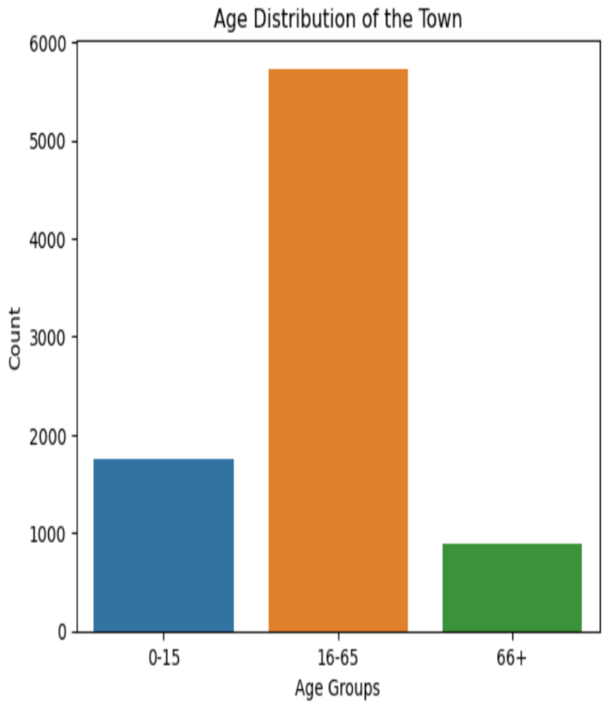
This population pyramid has eroded base and uneven distribution of ages along the labour force (15-64). The decrease between 5-9 year olds and 0-4 year olds suggest a general decrease in birth rate. This may be due to a combination of delay in marriage, a growing preference for smaller family size, among other factors. The sharp increase in the pyramid at 15-19 and 20-24 years (university age) suggests the influx of youthful migrants you are attending the university nearby. The fact that the population reduced disproportionately at the age group 25-29 (early career age), even reducing comparatively with 10-14 year olds might imply that the immigrants along with some of the young town natives typically leave the town after graduation to pursue opportunities elsewhere. This is followed by a continued increase in the population from ages 30-34 till 40-49 (mid to late career stage), with the most significant increase happening in the population of women ages 35-39 (mid career stage). This suggests a second influx of migrants especially women that are settling down in their careers, starting families, and seeking to maintain a work-life balance, a common trend in developed economies where women delay child bearing due to educational and career goals. Starting a family usually involves buying affordable houses, which may be obtainable in our hypothetical town, since it is close to two big cities.

The overall implication of this trend is that we have the majority of the population at late reproductive age (35 - 49 year olds), who will soon move to post reproductive age (50 and above), resulting in more retired people in the future. However, this group has managed to produce a lesser number of children; who are at their pre-reproductive age (0-14 years olds). This will result in less number of school-aged children in the future. In addition, the continued relocation of young residents at their early productive age (15-29) is worsening the situation. The population is shrinking, although the influx of young university students who are only temporary residents might make the problem look less obvious. The town's population will grow slower as the number of people reaching their reproductive age decreases.

B. The Dependency Ratio of the Population

To analyze the economic implication of the structure of the town’s population we have to calculate the dependency ratio which relates to the ratio of economically dependent residents as against the total labour force. The population will therefore be grouped into three:

- School Aged Children (0-15 years olds). The age when most are supposed to be in school
- Working Age Group (16 - 65 year olds). Legal age to work in the UK starts from 16 years
- The Elderly (66 and above). Although a forced retirement age of 65 no longer exists in the UK, UK state pension starts from 66 years.




```
THE AGE DISTRIBUTION OF THE TOWN:
School Aged Children; (0-15 yrs old):  1750
Working Age Group (16-65 yrs old):  5730
Elderly (66 yrs old and above):  897
```

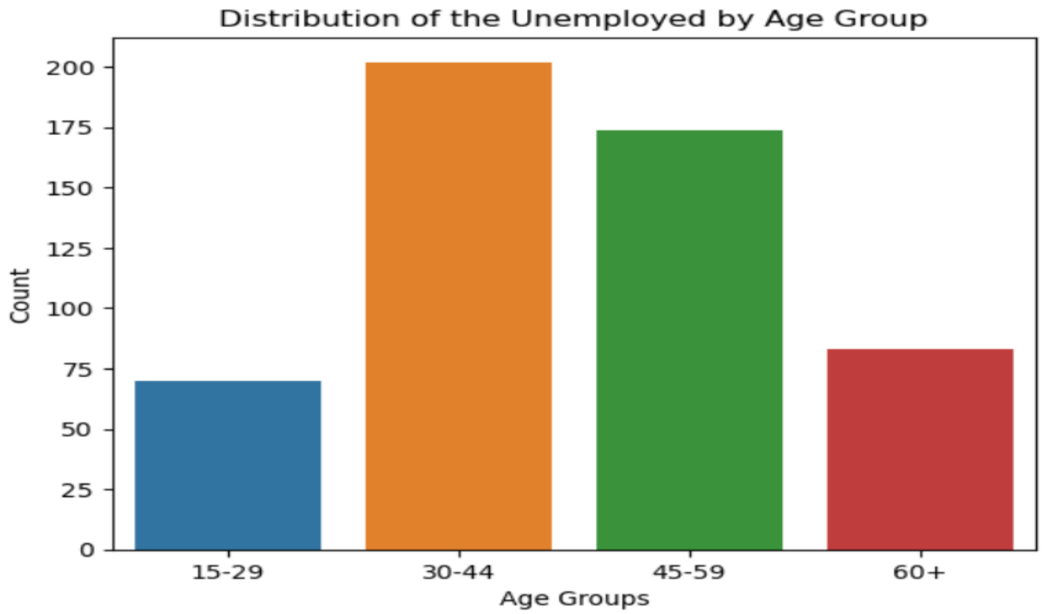
Dependency Ratio = $\frac{School\ Aged\ Children;\ (0-15\ yrs\ old)\ +\ Elderly\ (66\ yrs\ plus)}{Working\ Age\ Group\ (16-65\ yrs\ old)} \times 100$

$$= \frac{1750 + 897}{5730} \times 100 = 46.19\%$$

The ratio is expressed as the number of 'dependents' per 100 people aged 16-65 years. The population's dependency ratio is under 50% meaning that there are a sufficient number of working age groups to support the dependents.

2.2. Examining the Trend of Unemployment.

There is a total of 529 unemployed workers. A further analysis indicates that people who are in their mid careers (30-44) are most affected by unemployment. This is the when most are starting their families, the lack of job will greatly limit the capacity of these people to start a family



Rate of Unemployment
Though unemployment rate is minimal at 9.23% as calculated below

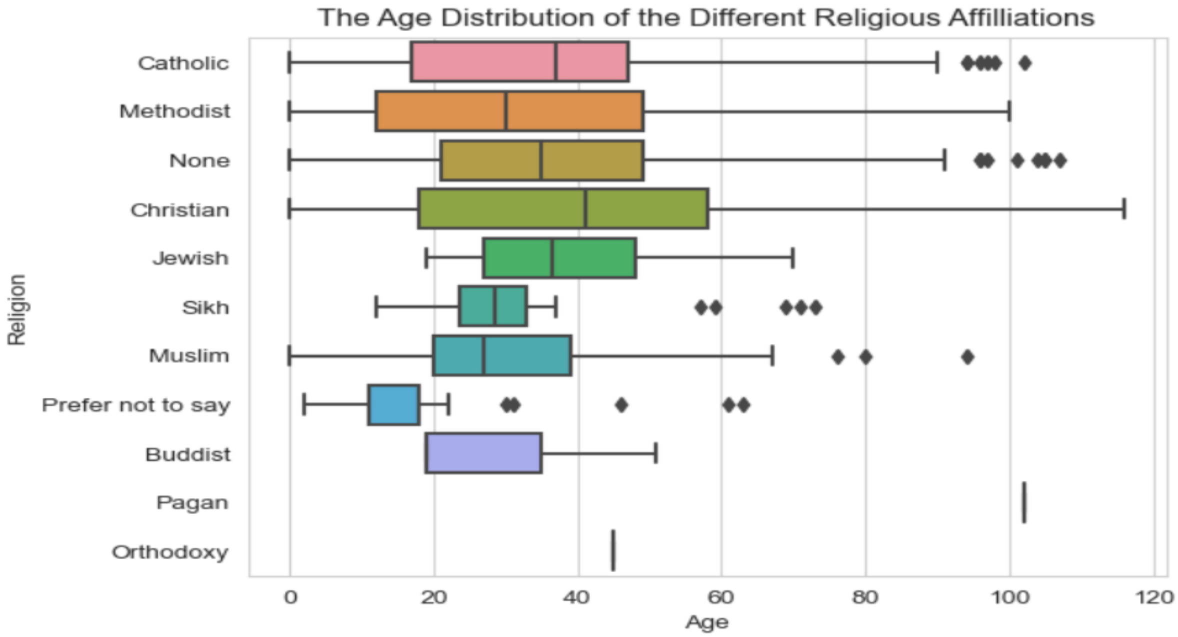
Unemployed Population = 529
Employed Population/Total Labour Force = 5730

Unemployment rate = $\frac{Unemployed\ Population}{Total\ Labour\ Force\ (Unemployed\ Population + Employed\ Population)} \times 100$

$$= \frac{529}{5730} \times 100 = 9.23\%$$

Rate of unemployment = 9.23%

2.3 Analyzing Religious Affiliations.

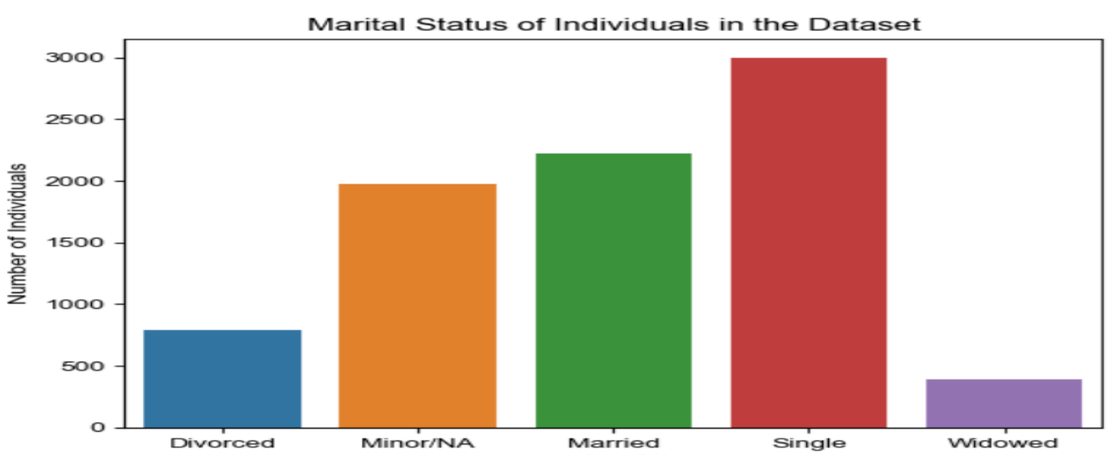


As detailed by the chart, the Christian religious group is the one that is growing the most. Those who declared non-affiliation have the greatest number of 0-20 year olds.

However, there is no need to use the land for any of the religious groups. Because, as indicated in the breakdown of the various religions, 42% of the population are not practicing any religion, combined with the Catholics who already have a church, you have 56%, a majority of the population who does not need any religious building. Another large group are the Christions; (2597). But how to decide the denomination of Christians to build a religious house for will be difficult.

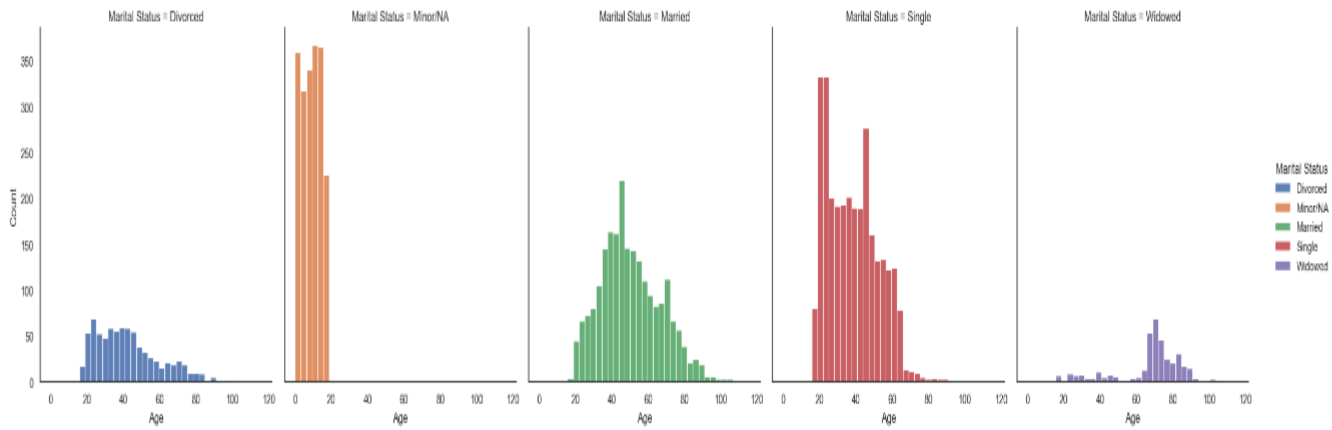
2.4 Examining Divorce and Marriage Rates and Its impact on Housing in the Town.

A simple chart revealing the marital status of the residents with a relatively low number of divorcees.



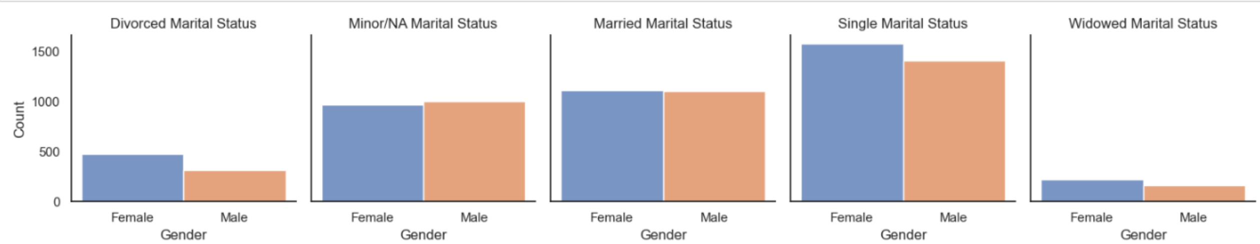
Age Distribution of Marital Status

```
sns.displot(x = 'Age', data = df, kind = 'hist', hue = 'Marital Status', col = 'Marital Status',alpha = 0.9);
```



Gender Distribution of Marital Status

```
# Plotting the Gender Distribution of Marital Status
sns.set_style('white')
g=sns.FacetGrid(df, col ='Marital Status', hue = 'Gender');
g.map(sns.histplot,'Gender');
g.set_titles(col_template = '{col_name} Marital Status');
```



The Divorce Rate

The rate of divorce is derived by calculating the crude divorce rate (CRM).

Crude Divorce Rate = $\frac{Number\ of\ Divorced}{Total\ Population} \times 1000$

$= \frac{791}{8,377} \times 1000 = 94$

= 94 per 1,000 people

The Impact of Divorce on Housing

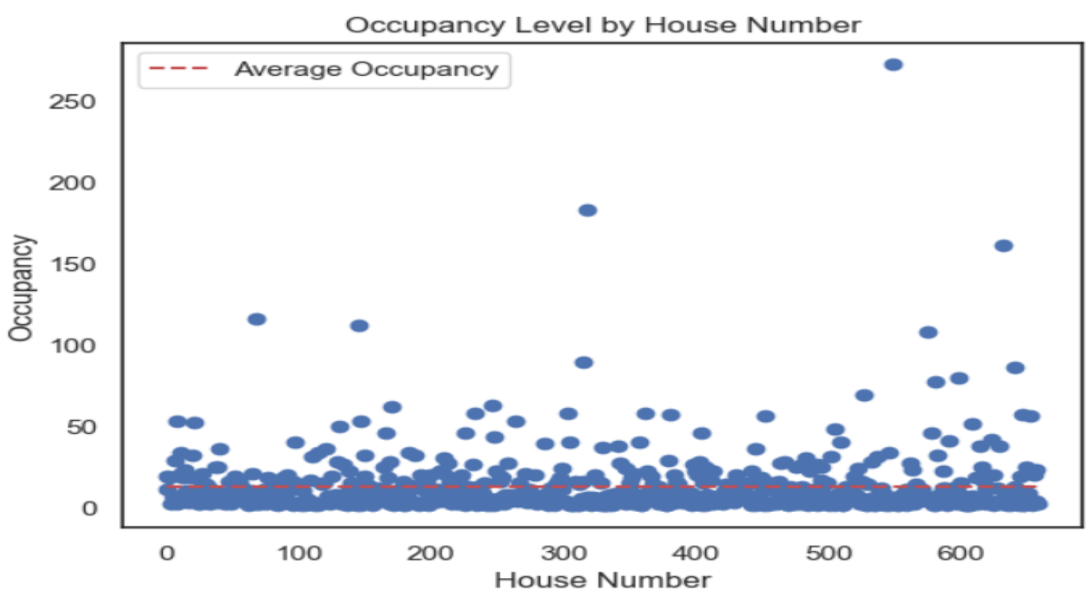
The demand for housing is linked to the formation of households, not necessarily population count. The formation of households is influenced by population growth and other factors like the age in which children move out of their parents home, marriage, cohabitation rates and divorce. Each divorce leads to the formation of a new household, although the total number of individuals remains constant, The split of households occasioned by divorce is also causing average household size to decrease significantly. At 94 per 1,000 people, though not significant, each divorce increase demand the housing market.

2.5 Examine the occupancy level

Calculating the Occupancy Level

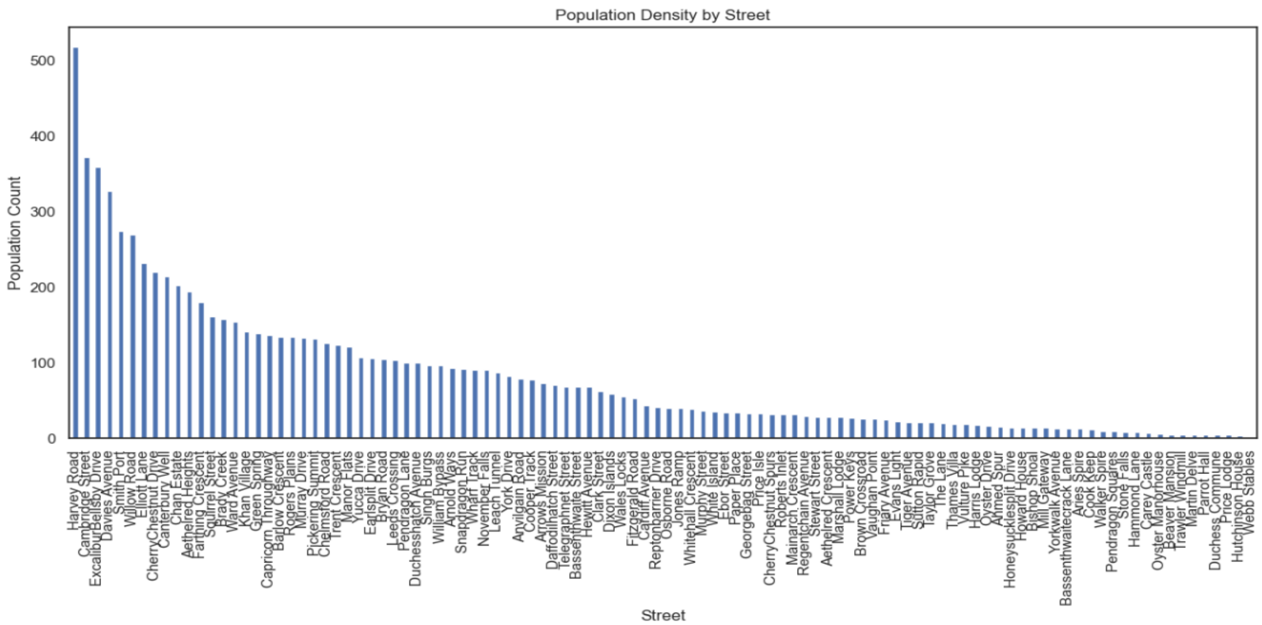
```
# Examine the occupancy level per house;


# Group the data by Surname and count the number of people in each house
household_size = df.groupby('Surname')['Street'].count()

# Calculate the average household size
avg_household_size = household_size.mean()

# Display the average household size
print(f"The average household size is {avg_household_size:.2f} people per house.")
```

```
The average household size is 12.71 people per house.
```



At 12.71 people per house, however, there is a low occupancy rate. Meaning that some houses are currently over used while others are scarcely occupied.

The graph below sheds more light on this



2.6 Analysing the Number of University Students and other Commuters in the Town

Identifying Commuters

To identify the total number of commuters, first I have to identify the total number of professionals in the town and since the town is a small town, it is safe to assume that the majority of these workers also commute from the town

Total Population = 8,377

Less students = 1584

Less Unemployed = 529

Less Children = 473

Less retired = 750

Balance = 5,041

Less University students = 533

Less PhD Students = 10

Total Number of workers living in the town = 4,493

Estimated 70% of this figure work outside the town,

Therefore 70% of 4,493 = 3,145 of workers who commute plus the number of University students (533) and PhD students (10) equals 3,688.

44% of the total population are commuters

2.7 What is the birth rate and death rate for the town?

To determine if new houses should be built, it is important to estimate how fast the population is growing or shrinking, this is done by calculating the birth and death rates for the town.

Calculating Birth Rate

```
# Calculating birth rate
new_birth = df['Age']==0
new_birth.sum()
```

```
89
```

```
# number of potential women of childbearing age.
Women_of_child_bearing_age = 0


for women in df['Gender']:
    if women == 'Female':
        Women_of_child_bearing_age += 1

print(Women_of_child_bearing_age)
```

```
4374
```

```
# Calculating birth rate
Birth_rate = round(89/4374*1000)
print('Birth_rate:', str(Birth_rate) + ' Per 1000')
```

```
Birth_rate: 20 Per 1000
```

Families with children are the key drivers of housing demand and increase in fertility rates can increase demand for housing. Birth rates within 10 to 20 per 1,000 is considered low. Therefore, the birth rate of 10 for this town should not be considered as a factor when making projections for new housing units.

Unfortunately, we can not make any analysis or projections on death rate because there's no information regarding deaths in the dataset.

3.0 RECOMMENDATION

What to Build

The town is small and cannot provide the economic platform for a labor force, but it has the advantage of being close to two other advanced economies. To retain and attract more of the labor force (15 to 64), for economic prosperity and future growth, the local government team needs to:

- Make it seamless to commute from the town to these two other economies.
- Provide enabling infrastructure to make leaving in the town comparable to the two big cities

The town should recognize advantage and concentrate on being a home to worker's in these two larger cities, this way it will attract more money in taxes and real estate.

Therefore, the local government should build a train station to take pressure off the roads and make commuting to the two larger cities very easy and convenient. The local government does not need to use the land for any of the religious group. As indicated in the breakdown of the various religions, 42% are not affiliated to any religious. It also difficult to decide on a particular religious group to build for.

What to Invest In

There is no evidence supporting a high rate of unemployment, therefore there is no need for investing in employment and training. On the contrary the town should invest in general infrastructure and services (waste collection; road maintenance, etc.) to make the town comparable to the two big cities, that way it will keep attracting commuters and reduce the number of young people who are forced to leave the town in search of better opportunities.

There seems to be need to invest in end of life care as the number of people within the 54-64 years, who are a considerable number in the population will soon move up the population pyramid. However, this will likely happen in the next decade, therefore it is not imminent,

References:

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